

Electronic supplementary material to:

How well do protected area networks support taxonomic and functional diversity in non-target taxa? The case of Iberian freshwaters

Guareschi S., Bilton D., Velasco J., Millán A., Abellán P.

APPENDIX A. Supplementary data and results.

Figure A1. Location of grid cells where “gap species” occur (i.e. those not represented in any protected area at a threshold of 50%).

Table A1. Functional traits used and their categories.

Table A2. Taxon affinity scores for each functional trait category.

Table A3. Comparison of taxonomic and functional α diversity between protected and unprotected grid cells at Iberian scale.

Table A4. Representation of taxonomic and functional γ diversity in protected area networks at Iberian scale for different protection thresholds.

Table A5. Representation of taxonomic and functional β diversity in protected area networks at Iberian scale across different protection thresholds.

Table A6. Comparison of taxonomic and functional α diversity between protected and unprotected grid cells at subregional scale (100-km-radius windows) for different protection thresholds.

Table A7. Representation of taxonomic and functional γ diversity in protected area networks at subregional scale (100-km-radius windows) for different protection thresholds.

Table A8. Representation of taxonomic β diversity in protected area networks at subregional scale (100-km-radius windows) for different protection thresholds.

Table A9. Representation of functional β diversity in protected area networks at subregional scale (100-km-radius windows) for different protection thresholds

Figure A1. Location of grid cells where “gap species” occur (i.e. those not represented in any protected area at a threshold of 50%). A, All species; B, vulnerable species.

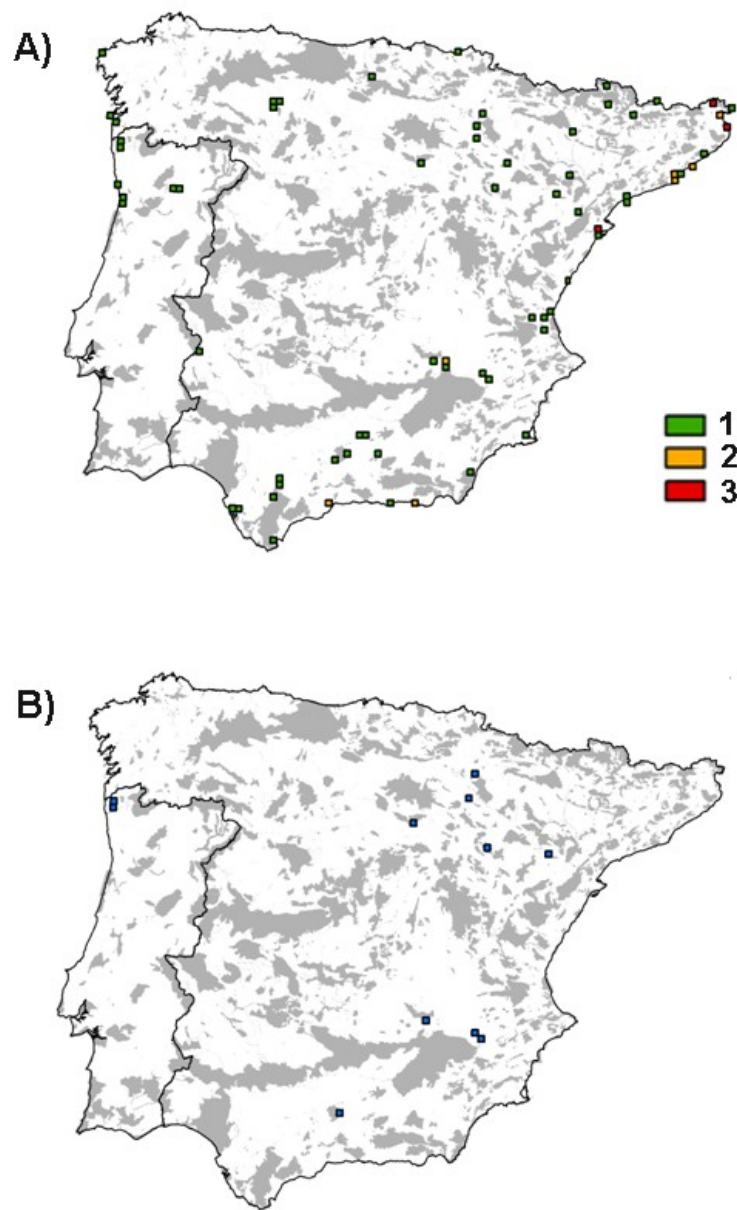


Table A1. Functional traits used and their categories.

Code	Biological traits	No	Categories
A	Maximal size	1	≤ 0.25 cm
		2	$> 0.25-0.5$ cm
		3	$> 0.5-1$ cm
		4	> 1 cm
B	Life cycle duration	1	≤ 1 year
		2	> 1 year
C	Potential number of reproduction cycles per year	1	≤ 1
		2	> 1
D	Aquatic stages	1	Egg
		2	Larva + pupa
		3	Adult
E	Reproduction	1	Clutches, cemented or fixed
		2	Clutches, in vegetation (endophytic)
		3	Clutches, terrestrial
F	Dispersal	1	Aquatic passive
		2	Aquatic active
		3	Aerial active
G	Resistance form	1	Diapause or dormancy
		2	None
H	Respiration	1	Tegument
		2	Gills
		3	Plastron
		4	Spiracle (aerial)
		5	Hydrostatic vesicle (aerial)
I	Locomotion and substrate relation	1	Flier
		2	Swimmer (surface swimmer included)
		3	Crawler
		4	Interstitial (endobenthic)
J	Food	1	Fine sediment + Microorganisms + Detritus < 1 mm
		2	Plant detritus ≤ 1 mm
		3	Living microphytes
		4	Living macrophytes
		5	Dead animals > 1 mm
		6	Living microinvertebrates
		7	Living macroinvertebrates
		8	Vertebrates
K	Feeding habits	1	Shredder
		2	Scraper
		3	Piercer (plants or animals)
		4	Predator (carver/engulfer/swallower)

Table A2. Taxa affinity scores for each functional trait category (see codes in Table A1).

Traits	A				B		C		D			E			F			G		H					I				J								K			
Modalities	1	2	3	4	1	2	1	2	1	2	3	1	2	3	1	2	3	1	2	1	2	3	4	5	1	2	3	4	1	2	3	4	5	6	7	8	1	2	3	4
Gyrinus	0	1	3	0	1	3	3	2	1	2	3	3	0	0	0	2	3	1	3	1	3	0	3	0	2	4	1	0	0	0	0	0	0	1	3	0	0	0	1	3
Aulonogyrus	0	0	3	0	1	3	0	3	1	2	3	3	0	0	0	2	3	1	3	0	2	0	3	0	1	4	1	0	0	0	0	0	0	1	3	0	0	0	1	3
Orectochilus	0	1	3	0	1	3	3	2	1	3	3	3	0	0	1	3	0	1	3	1	3	0	3	0	1	4	1	0	0	0	0	0	0	1	3	0	0	0	1	3
Peltodytes	0	3	0	0	1	3	1	3	1	2	3	0	3	0	0	1	3	0	3	0	2	0	3	0	2	2	3	0	0	0	2	3	0	1	0	0	0	3	2	1
Haliphus	0	3	0	0	1	3	1	3	1	2	3	0	3	0	0	3	2	1	3	0	2	0	3	0	2	2	3	0	0	0	2	3	0	1	0	0	0	3	2	1
Noterus	0	3	0	0	0	3	3	0	0	1	3	0	3	0	0	3	1	0	3	0	0	0	1	3	2	2	3	0	0	1	0	0	0	0	3	0	1	0	0	3
Hygrobia	0	0	3	0	0	3	3	0	1	3	3	0	3	0	0	0	3	0	3	0	0	0	3	3	2	3	1	0	0	3	0	0	0	0	3	0	0	0	0	3
Liopterus	0	0	3	0	0	3	3	1	1	1	3	0	3	0	0	1	3	1	3	1	0	0	1	3	2	3	2	0	0	0	0	0	0	1	3	0	1	0	0	3
Laccophilus	0	3	0	0	0	3	1	3	1	2	3	3	0	0	1	1	3	0	3	1	0	0	2	3	2	3	1	0	0	0	0	0	0	2	3	0	0	0	2	3
Hyphydrus	0	3	1	0	1	3	1	3	1	2	3	3	0	0	0	0	3	0	3	0	0	0	2	3	2	3	1	0	0	0	0	0	0	2	3	0	0	0	2	3
Hydrovatus	0	3	0	0	1	3	1	3	1	2	3	3	0	0	0	0	3	0	3	1	0	0	2	3	2	3	1	0	0	0	0	0	1	2	3	0	0	0	2	3
Yola	3	1	0	0	1	3	1	3	1	2	3	3	0	0	1	1	3	0	3	1	0	0	2	3	2	3	2	0	0	0	0	0	1	3	2	0	0	0	2	3
Bidessus	3	0	0	0	1	3	1	3	1	2	3	3	0	0	1	1	3	0	3	1	0	0	2	3	2	3	2	0	0	0	0	0	1	3	2	0	0	0	2	3
Hydroglyphus	3	0	0	0	0	3	0	3	1	2	3	3	0	0	0	1	3	0	3	0	0	0	2	3	3	3	2	0	0	0	0	0	1	3	2	0	0	0	2	3
Hygrotus	0	2	2	0	1	3	1	3	1	2	3	3	0	0	0	0	3	0	3	0	0	0	2	3	2	3	2	0	0	1	0	0	0	2	3	0	0	0	2	3
Herophydrus	0	3	0	0	1	3	1	3	1	2	3	3	0	0	0	0	3	0	3	0	0	0	2	3	2	3	2	0	0	1	0	0	0	2	3	0	0	0	2	3
Hydroporus	0	3	1	0	1	3	2	3	1	2	3	3	0	0	0	1	3	1	3	1	0	0	2	3	2	3	2	0	0	1	0	0	0	2	3	0	0	0	2	3
Graptodytes	0	3	0	0	1	3	1	3	1	2	3	3	0	0	1	1	3	0	3	1	0	0	2	3	2	3	2	0	1	0	0	0	0	2	3	0	0	0	2	3
Metaporus	0	3	0	0	1	3	1	3	1	2	3	3	0	0	0	1	3	0	3	0	0	0	2	3	2	3	2	0	1	0	0	0	0	2	3	0	0	0	2	3
Scarodytes	0	3	0	0	1	3	1	3	1	2	3	3	0	0	0	2	3	0	3	1	0	0	2	3	2	3	2	0	0	0	0	0	0	2	3	0	0	0	2	3
Stictonectes	0	3	1	0	1	3	1	3	1	2	3	3	0	0	1	2	3	1	3	1	0	0	2	3	1	3	2	0	0	0	0	0	0	2	3	0	0	0	2	3
Deronectes	0	1	3	0	1	3	1	3	1	2	3	3	0	0	2	1	3	1	3	1	0	0	2	3	1	3	2	0	0	0	0	0	0	2	3	0	0	0	2	3
Stictotarsus	0	0	3	0	1	3	1	3	1	2	3	3	0	0	0	1	3	0	3	0	0	0	2	3	1	3	2	0	0	0	0	0	0	2	3	0	0	0	2	3
Nebrioporus	0	0	3	0	1	3	1	3	1	2	3	3	0	0	1	2	3	1	3	1	0	0	2	3	1	3	2	0	0	1	0	0	0	2	3	0	0	0	2	3
Oreodytes	0	3	0	0	1	3	3	1	1	2	3	3	0	0	2	1	3	1	3	1	0	0	2	3	1	3	2	0	0	0	0	0	0	2	3	0	0	0	2	3
Platambus	0	0	3	0	0	3	3	1	1	2	3	3	0	0	2	1	3	1	3	1	0	0	2	3	1	3	2	0	0	0	0	0	0	1	3	1	0	0	2	3
Agabus	0	0	3	1	0	3	3	1	1	2	3	3	1	0	1	1	3	1	3	1	0	0	2	3	2	3	2	0	0	0	0	0	0	1	3	1	0	0	2	3
Ilybius	0	0	3	1	0	3	3	1	1	2	3	0	3	0	0	1	3	0	3	0	0	0	2	3	2	3	2	0	0	0	0	0	1	1	3	1	0	0	2	3
Rhantus	0	0	3	1	0	3	3	1	1	2	3	3	0	0	0	1	3	0	3	0	0	0	2	3	2	3	2	0	0	0	0	0	1	1	3	1	0	0	2	3

Colymbetes	0	0	0	3	0	3	3	1	1	2	3	3	0	0	0	1	3	0	3	0	0	0	2	3	2	3	1	0	0	0	0	0	1	1	3	1	0	0	2	3
Meladema	0	0	0	3	0	3	3	0	1	2	3	3	0	0	0	1	3	0	3	0	0	0	2	3	2	3	1	0	0	0	0	0	1	1	3	1	0	0	2	3
Eretes	0	0	0	3	0	3	3	0	1	2	3	3	0	0	0	0	3	0	3	0	0	0	2	3	3	3	1	0	0	0	0	0	1	1	3	1	0	0	2	3
Hydaticus	0	0	1	3	0	3	3	0	1	2	3	0	3	0	0	1	3	0	3	0	0	0	2	3	2	3	1	0	0	0	0	0	1	1	3	1	0	0	2	3
Graphoderus	0	0	0	3	0	3	3	0	1	2	3	0	3	0	0	1	3	0	3	0	0	0	2	3	2	3	1	0	0	0	0	0	1	1	3	1	0	0	2	3
Acilius	0	0	0	3	0	3	3	0	1	2	3	3	0	0	0	1	3	0	3	0	0	0	2	3	2	3	1	0	0	0	0	0	1	1	3	1	0	0	2	3
Dytiscus	0	0	0	3	0	3	3	0	1	3	2	0	3	0	0	1	3	0	3	0	0	0	2	3	2	3	1	0	0	0	0	0	0	0	3	3	0	0	3	2
Cybister	0	0	0	3	0	3	3	0	1	3	2	0	3	0	0	1	3	0	3	0	0	0	3	2	2	3	1	0	0	0	0	0	0	0	3	3	0	0	3	2
Helophorus	1	3	3	1	2	2	3	0	0	0	3	0	0	3	1	0	3	0	3	0	0	3	1	0	2	0	3	2	1	0	3	0	0	0	0	0	1	3	0	0
Hydrochus	0	3	0	0	1	3	3	1	0	0	3	0	0	3	1	0	3	0	3	0	0	3	1	0	2	0	3	2	1	0	3	0	0	0	0	0	1	3	0	0
Berosus	0	3	1	0	1	3	1	3	1	3	3	0	3	0	1	1	3	0	3	0	0	3	1	0	2	2	3	0	0	0	3	1	0	3	1	0	1	3	0	2
Hemisphaera	3	0	0	0	1	3	3	1	0	0	3	3	0	0	1	0	3	1	3	0	0	3	1	0	2	0	3	2	2	0	3	0	0	1	0	0	3	1	0	1
Chaetarthria	3	1	0	0	1	3	3	1	0	0	3	3	0	0	1	0	3	1	3	0	0	3	1	0	2	0	2	3	3	0	2	0	0	1	0	0	3	1	0	1
Paracymus	1	3	0	0	1	3	2	2	1	2	3	3	0	0	1	0	3	1	3	0	0	3	1	0	2	1	3	2	0	2	3	2	0	2	0	0	3	1	0	1
Anacaena	0	3	0	0	1	3	3	1	1	1	3	3	0	0	1	0	3	0	3	1	0	3	1	0	2	1	3	2	1	0	3	0	0	1	0	0	3	1	0	1
Laccobius	0	3	0	0	1	3	2	2	1	2	3	3	0	0	1	0	3	0	3	0	0	3	1	0	2	1	3	2	1	1	3	0	0	2	0	0	3	1	0	2
Helochares	0	3	0	0	1	3	1	3	2	2	3	3	0	0	1	0	3	0	3	0	0	3	1	0	2	1	3	2	0	1	1	3	0	0	2	0	1	3	0	2
Enochrus	0	1	3	0	1	3	1	3	1	2	3	3	0	0	1	0	3	0	3	0	0	3	2	0	2	1	3	1	0	1	1	3	0	0	2	0	1	3	0	2
Hydrobius	0	0	2	2	1	3	1	3	1	2	3	3	0	0	1	0	3	0	3	0	0	3	1	0	2	1	3	1	0	1	0	3	0	0	2	0	1	3	0	2
Limnoxenus	0	0	3	1	1	3	1	3	1	2	3	3	0	0	1	0	3	0	3	0	0	3	1	0	2	1	3	1	0	1	0	3	0	0	2	0	1	3	0	2
Hydrochara	0	0	0	3	1	3	1	3	1	2	3	3	0	0	1	0	3	0	3	0	0	3	1	0	2	3	2	0	0	1	0	3	0	0	2	0	0	3	0	2
Hydrophilus	0	0	0	3	1	3	1	3	1	2	3	3	0	0	1	1	3	0	3	0	0	3	1	0	2	3	1	0	0	1	0	3	0	0	2	0	0	3	0	2
Coelostoma	0	3	1	0	1	3	1	3	0	1	3	3	0	0	1	0	3	0	3	0	0	3	1	0	2	0	1	3	0	3	1	0	0	0	0	0	1	3	0	0
Hydraena	3	1	0	0	1	3	3	1	0	0	3	0	0	3	2	1	2	0	3	0	0	3	0	0	1	0	2	3	1	0	3	0	0	0	0	0	2	3	0	0
Limnebius	3	1	0	0	1	3	3	1	0	0	3	0	0	3	2	1	3	0	3	0	0	3	0	0	2	0	2	3	1	0	3	1	0	0	0	0	2	3	0	0
Aulacochthebius	3	0	0	0	1	3	1	3	0	1	3	0	0	3	1	0	3	0	3	0	0	3	0	0	2	0	2	3	2	1	3	0	0	0	0	0	2	3	0	0
Ochthebius	3	1	0	0	1	3	2	2	0	1	3	0	0	3	2	1	3	0	3	0	0	3	0	0	2	0	2	3	2	1	3	0	0	0	0	0	2	3	0	0
Potamophilus	0	0	3	0	1	3	3	0	0	3	0	0	3	0	1	0	3	1	3	1	3	1	0	0	1	0	3	1	0	3	3	0	0	0	0	0	3	1	0	0
Dupophilus	0	3	0	0	1	3	3	0	0	2	3	3	0	0	2	1	2	0	3	1	3	3	0	0	1	0	3	2	1	0	3	0	0	0	0	0	3	1	0	0
Stenelmis	0	1	3	0	1	3	3	0	0	2	3	3	0	0	2	1	2	0	3	1	3	3	0	0	1	0	2	3	1	0	3	0	0	0	0	0	2	3	0	0
Elmis	0	3	0	0	1	3	3	0	0	2	3	3	0	0	2	1	2	0	3	1	3	3	0	0	1	0	3	1	1	0	3	0	0	0	0	0	1	3	0	0
Esolus	1	3	0	0	1	3	3	0	0	3	2	3	0	0	2	1	2	0	3	1	3	3	0	0	1	0	3	1	1	0	3	0	0	0	0	0	1	3	0	0
Oulimnius	0	3	0	0	1	3	3	0	0	3	2	3	0	0	2	1	2	0	3	1	3	3	0	0	1	0	3	1	1	0	3	0	0	0	0	0	1	3	0	0

Limnius	0	3	0	0	1	3	3	0	0	2	3	3	0	0	2	1	2	0	3	1	3	3	0	0	1	0	3	1	1	0	3	0	0	0	0	0	1	3	0	0	
Normandia	0	3	0	0	1	3	3	0	0	2	3	3	0	0	2	1	2	0	3	1	3	3	0	0	1	0	3	1	1	0	3	0	0	0	0	0	1	3	0	0	
Riolus	0	3	0	0	1	3	3	0	0	2	3	3	0	0	2	1	2	0	3	1	3	3	0	0	1	0	3	1	1	0	3	0	0	0	0	0	1	3	0	0	
Dryops	0	3	1	0	1	3	3	0	0	2	3	0	1	3	1	1	3	0	3	3	0	3	3	0	2	0	3	1	0	3	3	0	0	0	0	0	3	1	0	0	
Pomatinus	0	0	3	0	0	3	3	0	0	1	3	0	0	3	0	1	3	0	3	3	0	3	3	0	1	0	3	2	0	0	3	3	0	0	0	0	3	3	0	0	
Iberoporus	3	0	0	0	1	3	1	3	1	2	3	3	0	0	1	3	0	0	3	1	0	0	2	3	0	3	0	0	0	0	0	0	0	0	3	3	0	0	0	3	3
Methles	0	3	0	0	1	3	1	3	1	2	3	3	0	0	0	2	3	0	3	0	0	0	2	3	2	3	0	0	0	0	0	0	0	0	0	3	0	0	0	3	3
Porhydrus	0	3	0	0	1	3	1	3	1	2	3	3	0	0	0	2	3	0	3	1	0	0	2	3	2	3	0	0	0	0	0	0	0	0	0	3	0	0	0	3	3
Rhitrodytes	3	0	0	0	1	3	3	1	1	2	3	3	0	0	2	1	3	1	3	1	0	0	2	3	1	3	0	0	0	0	0	0	0	0	0	3	0	0	0	3	3
Spaherius	3	0	0	0	1	3	3	1	1	1	3	3	0	0	1	1	3	1	3	0	0	0	2	3	1	0	3	1	0	0	3	0	0	0	0	0	3	0	0	0	
Trichonectes	0	3	0	0	1	3	1	3	1	2	3	3	0	0	1	2	3	0	3	0	0	0	2	3	2	3	0	0	0	0	0	0	0	0	0	3	0	0	0	3	3
Hydroscapha	3	0	0	0	1	3	3	1	1	1	3	3	0	0	1	1	3	1	3	0	0	0	2	3	1	0	3	1	0	0	3	0	0	0	3	0	3	0	0	0	
Boreonectes	0	3	0	0	1	3	1	3	1	2	3	3	0	0	0	2	3	0	3	0	0	0	2	3	2	3	0	0	0	0	0	0	0	0	0	0	0	0	0	3	3
Brychius	0	3	0	0	1	3	3	1	1	2	3	0	3	0	2	2	2	1	3	1	0	0	2	3	1	2	3	0	0	0	0	0	3	0	0	0	0	3	0	0	0
Canthydrus	0	3	0	0	0	3	1	3	0	1	3	0	3	0	0	2	3	1	3	0	0	0	2	3	2	3	0	0	0	1	0	0	0	0	3	0	0	0	3	3	
Chasmogenus	0	3	0	0	1	3	1	3	1	2	3	3	0	0	0	2	3	0	3	0	0	0	2	3	1	0	3	0	0	0	0	3	0	0	3	0	3	0	0	3	
Cymbiodyta	0	3	0	0	1	3	3	1	2	2	3	3	0	0	0	2	3	0	3	0	0	0	2	3	1	0	3	0	0	0	0	3	0	0	3	0	3	0	0	3	

Table A3. Comparison of taxonomic and functional α diversity (median $\pm$ SD) between protected and unprotected grid cells at Iberian scale. Significance was assessed by nonparametric Wilcoxon–Mann–Whitney tests. The number of cells (NC) considered as protected under each reserve network is also indicated.

	Threshold	NC	Protected	Unprotected	<i>W</i> -statistic	<i>P</i>
	d			d		
<i>Taxonomic diversity</i>						
RNAs		111				
	1	7	10 $\pm$ 17.0	6 $\pm$ 12.9	814626	< 0.001
	25	549	13 $\pm$ 17.5	6 $\pm$ 14.0	623307	< 0.001
	50	352	15 $\pm$ 17.9	6 $\pm$ 14.3	460201	< 0.001
	75	198	17 $\pm$ 19.4	7 $\pm$ 14.4	287963	< 0.001
	90	124	17.5 $\pm$ 19.6	7 $\pm$ 14.6	190224	< 0.001
N2000		181				
	1	7	9 $\pm$ 16.0	4 $\pm$ 10.6	638977	< 0.001
		106				
	25	4	12 $\pm$ 17.3	5 $\pm$ 12.2	895040	< 0.001
	50	690	13 $\pm$ 18.2	6 $\pm$ 13.0	751946	< 0.001
	75	412	16 $\pm$ 19.4	6 $\pm$ 13.5	540462	< 0.001
	90	252	16.5 $\pm$ 20.6	7 $\pm$ 14.0	358216	< 0.001
<i>Functional diversity</i>						
RNAs		111				
	1	7	0.268 $\pm$ 0.08	0.247 $\pm$ 0.09	530804	< 0.001
	25	549	0.270 $\pm$ 0.07	0.255 $\pm$ 0.09	387514	< 0.001
	50	352	0.271 $\pm$ 0.07	0.257 $\pm$ 0.09	287422	< 0.001
	75	198	0.273 $\pm$ 0.07	0.258 $\pm$ 0.09	180156	< 0.001
	90	124	0.273 $\pm$ 0.08	0.258 $\pm$ 0.09	120039	< 0.001
N2000		181				
	1	7	0.263 $\pm$ 0.08	0.232 $\pm$ 0.10	356280	< 0.001
		106				
	25	4	0.266 $\pm$ 0.07	0.251 $\pm$ 0.09	504469	< 0.001
	50	690	0.269 $\pm$ 0.07	0.255 $\pm$ 0.09	437074	< 0.001
	75	412	0.270 $\pm$ 0.07	0.256 $\pm$ 0.09	317764	< 0.001
	90	252	0.270 $\pm$ 0.08	0.258 $\pm$ 0.09	214261	0.003

Table A4. Representation of taxonomic and functional γ diversity in protected area networks at Iberian scale (protected) for different protection thresholds and comparison with the values expected from 1,000 random draws of an equal number of grid cells (random).

Network	Threshold	$\gamma_{\text{protected}}$	γ_{random}	<i>P</i>
<i>Taxonomic diversity</i>				
RNAs	1	477	463	< 0.001
	25	449	431	0.002
	50	419	404	0.040
	75	382	364	0.055
	90	351	326	0.037
N2000	1	484	478	0.001
	25	472	461	0.004
	50	458	443	0.002
	75	440	414	< 0.001
	90	405	382	0.012
<i>Functional diversity</i>				
RNAs	1	0.284	0.283	0.095
	25	0.281	0.283	0.920
	50	0.283	0.283	0.692
	75	0.282	0.283	0.646
	90	0.285	0.282	0.059
N2000	1	0.282	0.282	0.434
	25	0.282	0.283	0.765
	50	0.284	0.283	0.131
	75	0.285	0.283	0.063
	90	0.282	0.283	0.736

Table A5. Representation of taxonomic and functional beta diversity (β_{SOR}) and its decomposition in nestedness (β_{NES}) and turnover (β_{SIM}) in protected area networks (Protected) at Iberian scale across different protection thresholds and comparison with the values expected from 1,000 random draws of an equal number of grid cells (Random).

Network	Threshold	β_{SIM}			β_{NES}			β_{SOR}		
		Protected	Random	P	Protected	Random	P	Protected	Random	P
<i>Taxonomic diversity</i>										
RNAs	1	0.9952931	0.9952410	0.540	0.002809765	0.002888144	0.328	0.9981028	0.9981292	0.00 2
	25	0.9911104	0.9906279	0.064	0.005027810	0.005681436	0.008	0.9961382	0.9963094	0.00 2
	50	0.9866012	0.9858582	0.180	0.007379509	0.008561903	0.022	0.9939807	0.9944201	0.00 2
	75	0.9766507	0.9765143	0.970	0.012814420	0.014157220	0.274	0.9894651	0.9906715	0.00 2
	90	0.9652331	0.9654714	0.886	0.018580790	0.020751280	0.364	0.9838138	0.9862227	0.00 2
N2000	1	0.9971007	0.9970365	0.012	0.001724865	0.001799092	0.006	0.9988255	0.9988355	0.00 2
	25	0.9952859	0.9950127	< 0.001	0.002697699	0.003026270	0.002	0.9979836	0.9980389	0.00 2
	50	0.9928408	0.9924585	0.038	0.004044368	0.004568587	0.008	0.9968852	0.9970270	0.00 2
	75	0.9883894	0.9877729	0.176	0.006407057	0.007402056	0.008	0.9947964	0.9951749	0.00 2
	90	0.9812587	0.9809809	0.786	0.010412210	0.011472980	0.202	0.9916709	0.9924538	0.00 2
<i>Functional diversity</i>										
RNAs	1	0.9903451	0.9909953	0.002	0.00650441	0.00603762	0.002	0.9968495	0.9970330	0.00 2
	25	0.9795386	0.9819263	0.002	0.01368930	0.01207427	< 0.001	0.9932279	0.9940006	0.00 2
	50	0.9670745	0.9721515	0.002	0.02197082	0.01855651	<	0.9890454	0.9907080	0.00

							0.001			2
	75	0.9408807	0.9518428	0.002	0.03928814	0.03186513	0.008	0.9801688	0.9837079	0.002
	90	0.9049145	0.9262835	0.006	0.06300666	0.04826227	0.006	0.9679212	0.9745458	0.002
N2000	1	0.9670745	0.9721796	0.002	0.02197082	0.01853756	0.002	0.9890454	0.9907171	0.002
	25	0.9897171	0.9905612	0.002	0.00687048	0.00632402	< 0.001	0.9965876	0.9968852	0.002
	50	0.9834505	0.9855476	0.002	0.01113055	0.00966901	< 0.001	0.9945811	0.9952166	0.002
	75	0.9713965	0.9760498	0.002	0.01920146	0.01599141	0.002	0.9905979	0.9920412	0.002
	90	0.9529096	0.9616508	0.002	0.03179427	0.02546044	< 0.001	0.9847039	0.9871112	0.002

Table A6. Comparison of taxonomic and functional α diversity between protected and unprotected grid cells at subregional scale (100-km-radius windows) for different protection thresholds. It is shown the percentage of windows for which α diversity was significantly greater in protected than in unprotected grid cells.

Network	Threshold	% windows
<i>Taxonomic diversity</i>		
RNAs	1	44.8
	25	55.0
	50	50.3
	75	43.3
	90	38.0
N2000	1	55.7
	25	72.2
	50	68.0
	75	64.0
	90	52.7
<i>Functional diversity</i>		
RNAs	1	50.3
	25	29.4
	50	26.0
	75	20.9
	90	14.9
N2000	1	30.1
	25	33.2
	50	30.3
	75	23.6
	90	17.0

Table7. Representation of taxonomic and functional γ diversity in protected area networks at subregional scale (100-km-radius windows) for different protection thresholds. It is shown the percentage of windows for which γ diversity was significantly greater within protected areas than random.

Network	Threshold	% windows
<i>Taxonomic diversity</i>		
RNAs	1	26.3
	25	15.5
	50	14.1
	75	13.2
	90	11.9
N2000	1	19.8
	25	22.2
	50	17.1
	75	16.5
	90	18.2
<i>Functional diversity</i>		
RNAs	1	7.5
	25	5.0
	50	1.4
	75	3.5
	90	4.3
N2000	1	2.7
	25	2.1
	50	4.7
	75	1.4
	90	6.3

Table A8. Representation of taxonomic β diversity in protected area networks at subregional scale (100-km-radius windows) for different protection thresholds. It is shown the percentage of windows for which beta diversity (β_{SOR}) and its decomposition in nestedness (β_{NES}) and turnover (β_{SIM}) was significantly higher or lower among protected areas than random.

Network	Threshold d	β_{SIM}		β_{NES}		β_{SOR}	
		% higher	% lower	% higher	% lower	% higher	% lower
RNAs	1	0.34	12.96	8.98	1.72	1.05	44.88
	25	0.80	14.91	6.53	4.38	0.46	50.02
	50	2.13	19.57	4.04	4.30	0.04	61.30
	75	0.19	28.09	7.41	2.41	0.00	55.90
	90	0.00	26.06	9.72	0.00	0.00	47.05
N2000	1	0.76	2.31	0.42	2.18	0.13	42.79
	25	1.76	4.36	0.00	4.99	0.00	59.19
	50	0.09	14.77	3.59	0.60	0.00	66.20
	75	0.00	14.97	3.43	0.09	0.00	66.80
	90	0.55	21.90	6.93	0.46	0.55	61.83

Table A9. Representation of functional β diversity in protected area networks at subregional scale (100-km-radius windows) for different protection thresholds. It is shown the percentage of windows for which beta diversity (β_{SOR}) and its decomposition in nestedness (β_{NES}) and turnover (β_{SIM}) was significantly higher or lower among protected areas than random.

Network	Threshold d	β_{SIM}		β_{NES}		β_{SOR}	
		% higher	% lower	% higher	% lower	% higher	% lower
RNAs	1	0.00	68.62	0.00	8.98	0.00	79.49
	25	0.38	66.49	0.38	10.44	0.38	78.86
	50	0.00	30.39	14.96	1.17	0.00	50.91
	75	0.00	71.72	0.00	11.66	0.00	77.53
	90	0.00	70.76	0.00	6.52	0.00	67.09
N2000	1	NA	NA	NA	NA	NA	NA
	25	0.00	68.58	0.00	12.0	0.00	87.42
	50	0.34	36.83	24.16	1.32	0.00	56.94
	75	0.00	76.39	0.00	8.33	0.00	85.85
	90	0.00	71.03	0.00	13.17	0.00	78.28